DevOps assignment

📌 Overview

In this Assignment, you will be required to:

- provision an **AWS EKS cluster** using **Terraform**
- containerize a **Flask application** using **Docker**
- deploy the application using **Helm**
- automate the deployment using **GitHub Actions**.

📂 Assignment Steps

1️⃣ Provision AWS Infrastructure with Terraform

Write Terraform code to create the following AWS resources:

- 🏗️ **Infrastructure**:
  - An **S3 bucket** to store the Terraform state
  - A **VPC** with two subnets:
    - **Public subnet**
    - **Private subnet**
  - An **EKS cluster**
  - Two **EKS node groups** (one in each subnet)
  - An **AWS ECR** for storing container images
  - **IAM roles & necessary permissions**
- 📌 **Terraform minimum Outputs:**
  - EKS **cluster name**
  - EKS **cluster endpoint**
  - **ECR repository URL**
  - **IAM role ARN**
  - please add any other relevant outputs
- 📖 **Please provide a README explaining how to deploy the infrastructure**

2️⃣ Dockerize the Provided Application

- In the app/ directory, you will find a **Flask application (app.py)**.
- Write a **Dockerfile** to containerize it.
- Build the **Docker image** and push it to the **AWS ECR** repository created in the previous step.
- 📖 **Provide a README explaining how to build and push the image**.

3️⃣ Deploy the Application Using Helm

- Write a **Helm chart** to deploy the application to **EKS**.
- The application should be **accessible from the internet** after deployment.
- Provide the **public endpoint URL** for accessing the application.

4️⃣ Automate Deployment with GitHub Actions

Set up a **GitHub Actions workflow** to:

1. **Build & Push Docker Image** to ECR on **push to main branch**.
2. **Deploy/Update the Application** in EKS using **Helm**.

Summary

Your repository should contain:
✅ Terraform code for AWS infrastructure setup
✅ Dockerfile for containerizing the app
✅ Helm chart for deploying the app to EKS
✅ GitHub Actions workflow
✅ A **README.md** explaining how to set up and execute the project
✅ please use best practices, performace, cost and security aware methodologies wherever possible.
🎯 Bonus 1 - write a github actions workflow for deploying the infrastructure using terraform.
🎯 Bonus 2 - deploy a monitoring for the application.

🍀Good Luck !